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Atomic layer deposition of ternary ruthenates by combining metalorganic precursors with RuO₄ as the co-reactant†

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In this work, the use of ruthenium tetroxide (RuO₄) as a co-reactant for atomic layer deposition (ALD) is reported. The role of RuO₄ as a co-reactant is twofold: it acts both as an oxidizing agent and as a Ru source. It is demonstrated that ALD of a ternary Ru-containing metal oxide (*i.e.* a metal ruthenate) can be achieved by combining a metalorganic precursor with RuO₄ in a two-step process. RuO₄ is proposed to combust the organic ligands of the adsorbed precursor molecules while also binding RuO₂ to the surface. As a proof of concept two metal ruthenate processes are developed: one for aluminum ruthenate, by combining trimethylaluminum (TMA) with RuO₄; and one for platinum ruthenate, by combining MeCpPtMe₃ with RuO₄. Both processes exhibit self-limiting surface reactions and linear growth as a function of the number of ALD cycles. The observed saturated growth rates are relatively high compared to what is usually the case for ALD. At 100 °C sample temperature, growth rates of 0.86 nm per cycle and 0.52 nm per cycle are observed for the aluminum and platinum ruthenate processes, respectively. The TMA/RuO₄ process results in a 1 : 1 Al to Ru ratio, while the MeCpPtMe₃/RuO₄ process yields a highly Ru-rich composition with respect to Pt. Carbon, hydrogen and fluorine impurities are present in the thin films with different relative amounts for the two investigated processes. For both processes, the as-deposited films are amorphous.

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Introduction

In the past decades, atomic layer deposition (ALD) has become an indispensable tool for nanotechnology.^{1,2} It relies on self-limiting chemical reactions between a sample surface and gas-phase precursors, which enables the synthesis of inorganic materials with atomic precision. ALD is traditionally achieved by alternating metal precursor (*e.g.* a metalorganic compound) and co-reactant (*e.g.* H₂O or O₂ gas/plasma) in a two-step process. In ALD of elemental thin films, in a simplified view the metal precursor first binds to the surface through saturated chemisorption, after which the co-reactant removes all of the ligands from the surface except for the intended element itself. So far, ALD of elemental films has been demonstrated almost exclusively for noble metals. For example, ALD of Pt can be achieved by combining the MeCpPtMe₃ precursor with O₂ at substrate temperatures above 150 °C, or ozone at lower

substrate temperatures.^{3,4} In these processes, the co-reactant oxidizes the ligands of the adsorbed precursor molecules, and further oxidizes the surface in a self-limiting manner, producing a surface oxide that is easily reducible. It oxidizes the ligands of incoming precursor molecules during the next precursor pulse, liberating electrons to reduce the metal center.⁵ Similar combustion chemistries have been reported for ALD of Ru combining a metalorganic precursor with O₂.⁶ Alternatively, Ru ALD can be achieved using the inorganic RuO₄ precursor in combination with H₂ gas or plasma as co-reactant at 100 °C.^{7,8} The RuO₄ precursor oxidizes the Ru surface of the growing film, while at the same time depositing RuO₂ in a saturated manner. During the H₂ pulse the surface is reduced back to metallic Ru.

In ALD of binary compounds, such as metal oxides or metal nitrides, next to the removal of the ligands of the adsorbed precursor molecules, the co-reactant often also supplies the second element (*e.g.* the oxygen or nitrogen atoms). A prototypical process is thermal ALD of Al₂O₃, where the Al species are supplied by Al(CH₃)₃ (trimethylaluminum or TMA) and the O species by H₂O. The surface chemistry in this case uses self-limiting ligand exchange reactions in an acid-base chemistry.^{5,9}

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Next to elemental and binary materials, the deposition of ternary and quaternary materials by ALD has attracted a lot of attention.¹⁰ The most common way to deposit ternary or quaternary materials is to combine multiple binary ALD processes into supercycles. Considering growth of ternary ruthenium oxides, Cheon *et al.* demonstrated ALD of RuAlO at 230 °C by combining ALD of Al₂O₃ and ALD of Ru in an alternating fashion, yielding a nanocomposite.¹¹ The Al₂O₃ ALD subcycles consisted of pulses of TMA and H₂O, while the Ru ALD subcycles consisted of pulses of IMBCHRu [(η⁶-1-isopropyl-4-methylbenzene)(η⁴-cyclohexa-1,3-diene)Ru(0), C₁₅H₂₂Ru] and O₂ gas. The microstructure of the resulting films comprised Ru nanocrystals separated by amorphous Al₂O₃, displaying enhanced Cu diffusion barrier properties compared to the columnar microstructure typically achieved for pure Ru ALD films. Han *et al.* demonstrated SrRuO₃ growth at 230 °C by combining ALD of SrO, using the Sr(iPr₃Cp)₂ precursor with O₂ gas, with the thermal decomposition of RuO₄ to RuO₂ on the SrO surface.¹² Note, however, that the latter is not a self-saturating reaction. Next to ternary ruthenium oxides, ternary ruthenium nitrides have been deposited by ALD, also by using the supercycle approach.^{13,14}

The supercycle approach in principle offers excellent composition control by tuning the fraction of each binary process within the supercycle, but several challenges are encountered.¹⁰ As ALD processes for different materials are alternated, overlapping ALD temperature windows are required to achieve saturated conditions and nucleation effects of one material on the other might occur, which can be difficult to control. In the specific case of Ru-containing oxides, the use of strong oxidants such as O₂ plasma in the sequence can pose challenges because it can lead to etching of deposited Ru species as volatile RuO₄,¹⁵ impacting composition and morphology control. Finally, if two successive binary materials do not mix well, one might end up with a multilayer or mixture of two separate material phases.

Alternative approaches to deposit ternary materials by ALD include co-dosing of precursors and use of a multiconstituent precursor. Such a precursor has multiple metal centres and therefore several elements are introduced on the surface during the same precursor pulse.¹⁶ The co-reactant can also be multiconstituent, which is for example the case when it acts as oxidizing agent and metal source, as demonstrated before for metal alkoxides.¹⁷ Rimoldi *et al.* reported the growth of rhenium aluminium oxide thin films by alternating TMA and ReO₃Me.¹⁸ This process was later investigated in detail by Gebhard *et al.*¹⁹ They reported the growth of ReAl₂O₃CH₃ with an exceptionally high growth rate of 0.45 nm per cycle at 150 °C. The mechanistic studies revealed that catalytic chemistry at the Re center occurs during ALD, yielding hydrogen and olefins as reaction products. To the best of our knowledge, routes based on multiconstituent reactants have not been reported for ALD of ternary ruthenium oxides.

In this work, we will demonstrate that RuO₄ can be used as a multiconstituent co-reactant in a two-step ALD process for the growth of metal ruthenates at substrate temperatures

below 125 °C. In this ALD-route, RuO₄ acts as both the oxidizing agent and the Ru-source. As a proof of concept, we developed two-step ALD processes for aluminum and platinum ruthenate, by combining trimethylaluminum (TMA) or (methylcyclopenta-dienyl)trimethylplatinum (MeCpPtMe₃) with RuO₄, respectively. As metal ruthenates present an intriguing class of fundamentally and technologically relevant ternary oxides,^{20–22} due to their interesting electrical and magnetic properties, and excellent thermal and chemical stability, the presented method is expected to create opportunities for fields in search of conformally deposited ultrathin ruthenate thin films, for example, as solid-state (electro)catalyst or electrode materials.

Experimental

TMA/RuO₄ process

The ALD-reactor used for the aluminum ruthenate depositions is a home-built vacuum system, equipped with a turbomolecular pump and an oil-based backing pump, allowing the system to achieve a base pressure of 10^{−7} mbar. The used precursors for the aluminum ruthenate depositions were trimethylaluminum (TMA, Strem Chemicals, min. 98%) and RuO₄ (ToRuSTM: solution of RuO₄ in an inert organic solvent, Air Liquide). Both TMA and RuO₄ were stored in stainless steel vessels at room temperature, and were pulsed into the chamber by vapor draw through stainless steel tubing and pneumatically controlled inlets. The tubing was heated to 35 °C for both TMA and RuO₄. The pressure in the ALD chamber during the TMA and RuO₄ pulsing was set to 0.005 mbar and 0.02 mbar, respectively, by using needle valves at the precursor inlets.

MeCpPtMe₃/RuO₄ process

The ALD-reactor used for the platinum ruthenate depositions is a home-built vacuum system, allowing for synchrotron-based *in situ* XRF and GISAXS measurements.²³ The vacuum chamber is equipped with a turbomolecular pump backed by an oil-based rotation pump, allowing the system to reach a base pressure of 10^{−6} mbar. The two-step process is performed by alternating exposures of (methylcyclopentadienyl)trimethylplatinum (MeCpPtMe₃) (Strem Chemicals, 99%) and RuO₄ (ToRuSTM: solution of RuO₄ in an inert organic solvent, Air Liquide). Both precursors are vapor drawn into the ALD reactor through heated stainless steel tubing. For the Pt precursor Ar carrier gas was used, while no carrier gas was used for ToRuS. The Pt precursor is kept in a glass vessel at 35 °C, with the tubing at 45 °C. The RuO₄ precursor is kept in a stainless steel vessel at room temperature (no heating), with the tubing at 50 °C. The precursor pulsing in this process was of a static nature. This means that during each pulse, the gate valve between the ALD chamber and the turbo pump was closed, the precursors were injected for 6 s until a high pressure was achieved (Pt: 1 mbar, Ru: 2 mbar). After 9 s at this “static” pressure (the total pulse time is therefore 15 s), the chamber

was pumped down to base pressure again by first bypassing the chamber to the rotary pump, before opening the gate valve towards the turbo pump. In the three-step ALD process, static H_2 (20% H_2 in Ar) with a pressure of 4 mbar is added following these two precursor pulses. In this case, the injection time for H_2 was 8 s and the time at 4 mbar static pressure 20 s, giving a total pulse time of 28 s.

In situ characterization during the $MeCpPtMe_3/RuO_4$ process

In situ XRF measurements were performed at the DUBBLE BM26B beamline of the ESRF synchrotron. The samples were irradiated by a 11.98 kV beam at an incidence angle of 0.5° in between ALD cycles, and the fluorescent X-rays were detected by a Vortex silicon drift detector. From the XRF data, both *in situ* and *ex situ* (see below), an equivalent thickness of Pt and Ru was extracted. This was done by first constructing calibration curves of the Pt M and Ru L peak intensities *versus* the XRR thickness of sputtered Pt and Ru films, respectively, and afterwards using these calibration curves to extract an equivalent Pt or Ru thickness of the ALD films based on the XRF data.

Thin film characterization

X-ray reflectometry (XRR) and X-ray diffraction (XRD) measurements were performed on a Bruker D8 diffractometer using $Cu\ K\alpha$ radiation. The film thickness and density were determined by fitting simulated XRR patterns to the acquired reflectometry data. *Ex situ* X-ray fluorescence spectroscopy (XRF) was performed using a Bruker ARTAX μ -XRF spectrometer. Scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDX) was performed on an FEI Quanta 200F instrument. EDX spectra were acquired by scanning a 5 kV electron beam onto a large surface area, and using an EDAX silicon-drift detector. X-ray photoelectron spectroscopy (XPS) was performed on a Thermo ScientificTM Theta Probe instrument using monochromated Al $K\alpha$ X-rays generated at 15 kV, and focused into a 300 μm spot by the Si crystal monochromator mirror. The concentric hemispherical analyzer was set to a pass energy of 200 eV. Depth profiling was performed using an EX05 Ar^+ ion gun. Time-of-flight elastic recoil detection analysis (TOF-ERDA) was performed using 8 MeV $^{35}Cl^{4+}$ incident ions from the 1.7 MV Pelletron accelerator. The TOF-ERDA data were analysed by means of Potku software.

Results and discussion

TMA/ RuO_4 process

As a first illustration of the use of RuO_4 as a co-reactant, it was combined with trimethylaluminum (TMA) in a two-step process. Six different samples were prepared at different ALD cycle counts, and Fig. 1a displays their film thickness determined by fitting simulated patterns to XRR data. It is clear that the TMA/ RuO_4 process displays linear growth with negligible growth delay. By fitting a straight curve to this set of data points using the least-squares method, a growth per cycle

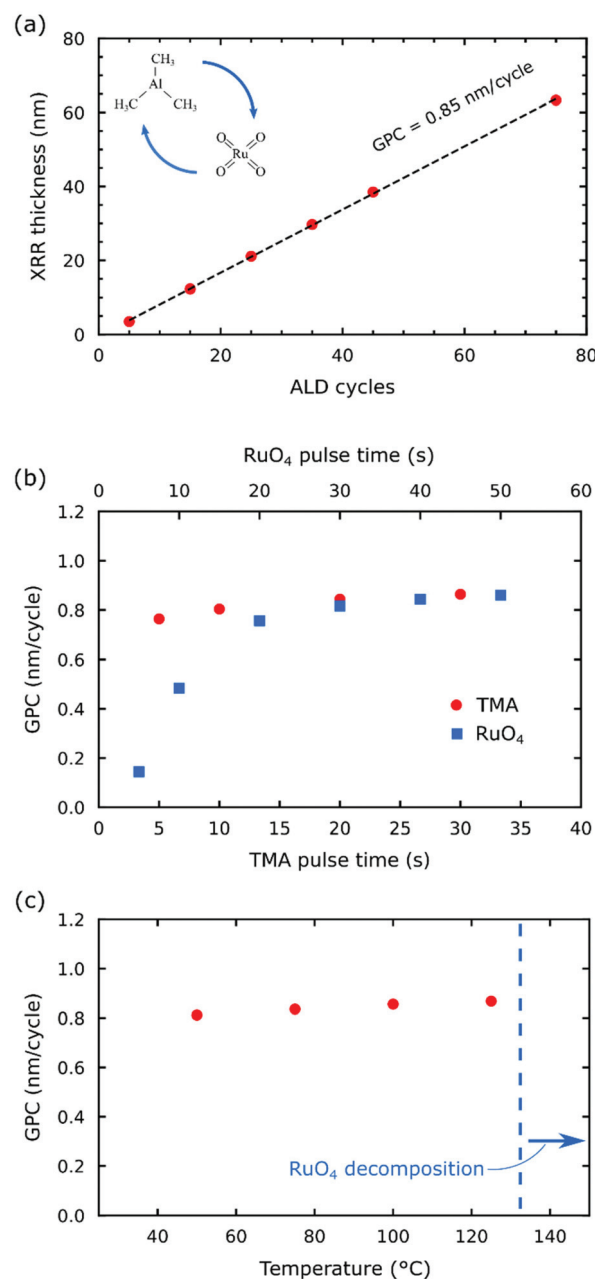


Fig. 1 (a) Growth curve of the TMA/ RuO_4 process at 100 °C sample temperature, showing the film thickness as a function of the number of ALD cycles on a SiO_2 substrate. The TMA and RuO_4 pulse times were set to 20 s and 40 s, respectively. (b) Growth per cycle determined by *ex situ* XRR as a function of the precursor (TMA) pulse time for a fixed co-reactant (RuO_4) pulse time of 40 s [red circles, bottom x-axis] and as a function of the co-reactant (RuO_4) pulse time for a fixed precursor (TMA) pulse time of 20 s [blue squares, top x-axis], both at 100 °C sample temperature. (c) Growth per cycle determined by *ex situ* XRR as a function of the substrate temperature for the saturated TMA/ RuO_4 -process.

(GPC) of 0.85 nm per cycle was extracted. This is a high GPC for ALD, as the growth rates are typically below 0.1 nm per cycle. From Fig. 1b, it is obvious that the GPC saturates with both the TMA and the RuO_4 pulsing time, which is one of the

basic conditions for ALD. In what follows, TMA pulse times of 20 s and RuO_4 pulse times of 40 s were selected as saturated process conditions. The influence of the sample temperature on the growth rate was also investigated. As the RuO_4 -precursor is known to thermally decompose for temperatures above 125 °C,⁷ only temperatures lower than this decomposition limit were evaluated. From Fig. 1c it is evident that the temperature window of the process is relatively flat, with a GPC going from 0.81 nm per cycle at 50 °C to 0.87 nm per cycle at 125 °C.

The XRR pattern obtained for the 75 cycle sample is displayed in Fig. 2a and is consistent with a smooth film of 64.1 nm deposited onto SiO_2 , with a density of 5.1 g cm^{-3} . This value lies in between the typical density of ALD-grown amorphous Al_2O_3 thin films, *ca.* 3.0 g cm^{-3} ,²⁴ and the tabulated density of crystalline RuO_2 , 6.97 g cm^{-3} . The same sample was characterized by X-ray diffraction (XRD), but no peaks were observed in the diffractogram, suggesting that the films deposited by the TMA/ RuO_4 -process are amorphous. Next, we characterized the sample by scanning electron microscopy (SEM). Consistent with the XRR data, the secondary electron images were featureless, suggesting that the substrate was uniformly coated with a flat ALD film. Energy-dispersive X-ray spectroscopy (EDX) was performed by scanning a 5 kV electron beam onto a large surface area, and the obtained spectrum is shown in Fig. 2b. Based on the Al K and Ru L

signals, quantification of the Al and Ru atomic concentration was carried out using ZAF correction to obtain the Al : Ru ratio. This analysis suggests that an equal amount of Al and Ru atoms are present in the deposited film. Quantification of the carbon content in the films was achieved *via* elastic recoil detection (ERD) analysis. Fig. 3 shows the extracted depth profile. In line with the EDX result, ERD yielded a 1 : 1 ratio of Al to Ru in the film with an atomic concentration of 16.0 ± 0.6 at% Al and 17.3 ± 0.6 at% Ru. The O concentration was 48.1 ± 1.3 at%. Carbon as well as hydrogen can be seen throughout the sample at a similar concentration of $\sim 9.0\%$ (9.3 ± 0.6 at% C and 8.7 ± 0.6 at% H), yielding an approximate film stoichiometry of $\text{RuAlO}_3\text{C}_{0.5}\text{H}_{0.5}$. This result suggests that the effectiveness of RuO_4 to combust the methyl ligands of the TMA molecules at 100 °C is not 100%, but that some fragments of ligands are incorporated in the thin film as carbon and hydrogen contamination. This is not uncommon in low temperature ALD surface chemistries, and is, for example, also seen in the thermal TMA/ H_2O process.²⁴ Trace amounts of fluorine (0.6 ± 0.15 at%) are also detected in ERD, originating from the solvent used to dissolve the RuO_4 precursor. Note that the elemental composition of the film was also investigated by X-ray photoelectron spectroscopy (XPS). However, due to the exact overlap between the C 1s and the Ru $3d^{3/2}$ peak and the complexity of the involved asymmetric line shapes,²⁵ it was not possible to determine the carbon content from the XPS data. The relative concentrations determined based on the Al 2p, O 1s and Ru 3p signals detected in the bulk of the film (Fig. S1, ESI†), *i.e.* after removing material from the top of the film by sputtering, strongly deviated from the EDX and ERD results, pointing to quantification challenges in XPS,²⁶ potentially due to deviations between actual and theoretical relative sensitivity factors, or sputter-induced changes in elemental distribution.

In summary, alternating TMA and RuO_4 in a two-step process yielded saturated growth during both surface reactions, pointing towards an ALD chemistry. Benefits are the low deposition temperature (<125 °C) and the high saturated GPC of 0.85 nm, allowing for rapid deposition of amorphous aluminium ruthenate thin films with an Al : Ru ratio of 1 : 1. These observations suggest that a relatively large and similar amount of TMA and RuO_4 molecules can react with the surface during each exposure. We propose that RuO_4 combusts

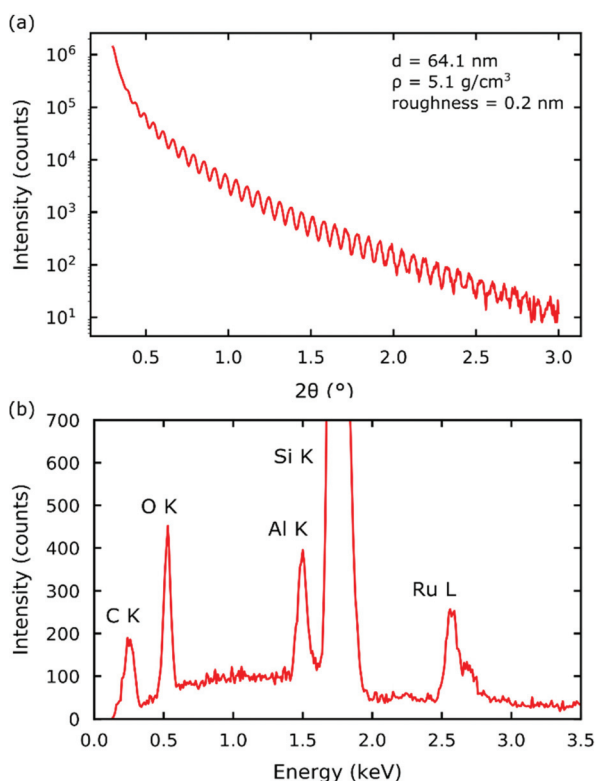


Fig. 2 (a) XRR data acquired for the 75 cycle sample from Fig. 1a. The XRR fit parameters are also indicated. (b) EDX spectrum with identified peaks.

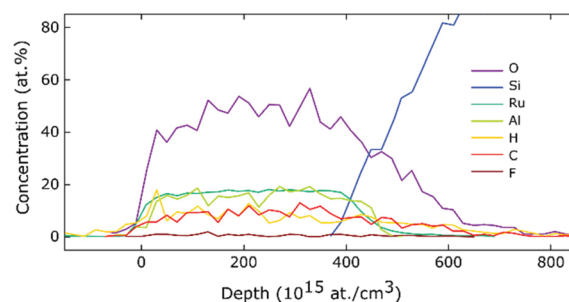


Fig. 3 Elemental distribution obtained from analysis of ERD spectra acquired for a *ca.* 50 nm film grown with the TMA/ RuO_4 ALD process.

the methyl ligands of adsorbed TMA molecules, while leaving RuO_2 on the surface.²⁷ Tallarida *et al.* studied the reaction of TMA with Ru and RuO_2 surfaces by *in vacuo* synchrotron-based XPS. They reported an enhanced initial Al_2O_3 growth on RuO_2 and concluded that reaction with TMA leads to a reduction of RuO_2 on the surface to Ru.²⁸ In the present TMA/ RuO_4 process, the reduced Ru on the surface after the TMA step may cause in turn enhanced deposition during the RuO_4 step, by re-oxidation of the surface Ru to RuO_2 and concomitant deposition of RuO_2 , as in the reported RuO_4/H_2 ALD process.⁷ Such

a surface chemistry could potentially explain the high saturated GPC.

MeCpPtMe₃/RuO₄ process

To illustrate the generality of the presented approach, platinum ruthenates were deposited as well, combining MeCpPtMe₃ and RuO_4 . The growth was carried out in a dedicated ALD setup enabling synchrotron-based *in situ* characterization of the process at the DUBBLE BM26 beamline of the ESRF. Fig. 4a displays the growth curve obtained from *in situ* X-ray fluorescence (XRF) measurements, and reveals that linear growth is achieved after a few cycles of non-linear growth at the start of the process. It is clear that the number of Ru atoms (right y-axis) deposited in each cycle is larger than the number of Pt atoms (left y-axis). Importantly, the Pt to Ru ratio seems to be relatively stable during film growth. The *ex situ* XRR data acquired on this 50 cycle sample are shown in Fig. 4b. The pattern is consistent with a uniform, low-roughness film of 25.9 nm deposited onto Si with a density of approximately 6 g cm^{-3} . This value lies closer to the bulk density of RuO_2 , 6.97 g cm^{-3} , than that of PtO_2 , 10.5 g cm^{-3} , in line with a Ru-rich composition. Based on the geometrical layer thickness as determined by XRR, we can estimate the growth rate of the process at 0.52 nm per cycle.

The saturation behaviour of the MeCpPtMe₃/RuO₄ process was investigated, confirming saturated growth for a 15 s MeCpPtMe₃ exposure time (Fig. 5a). The growth rate of the MeCpPtMe₃/RuO₄ process as a function of RuO_4 exposure is shown in Fig. 5b. In this case, the increase in GPC clearly levels off at 45 s RuO_4 exposure. As this corresponds to three successive static RuO_4 pulses of 15 s, and this is unfavorable towards the time efficiency of the process and the precursor consumption, a single 15 s RuO_4 pulse was selected for further experiments. In this way, it was found that the growth rate of the MeCpPtMe₃/RuO₄ process increases with substrate temperature, from 0.26 nm per cycle at 40 °C to 0.54 nm per cycle at 120 °C (Fig. 5c).

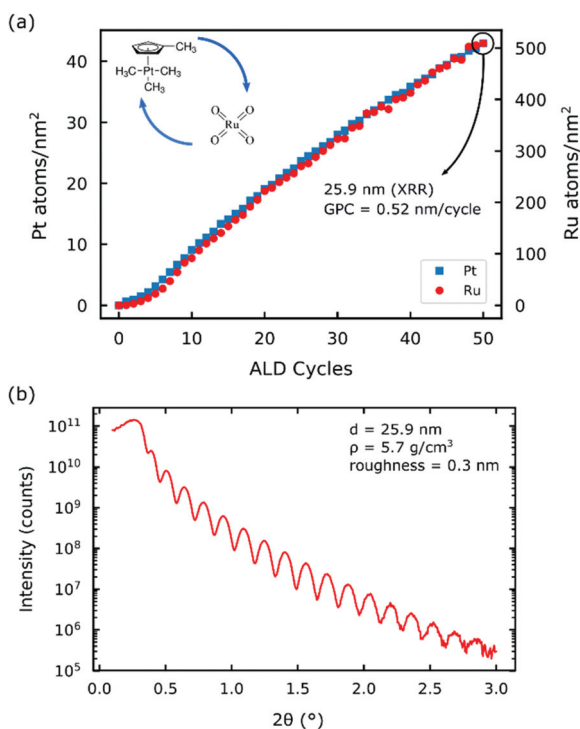


Fig. 4 (a) *In situ* XRF growth curve of the MeCpPtMe₃/RuO₄ process at 120 °C. (b) *Ex situ* XRR data acquired on the sample in (a) after 50 ALD cycles.

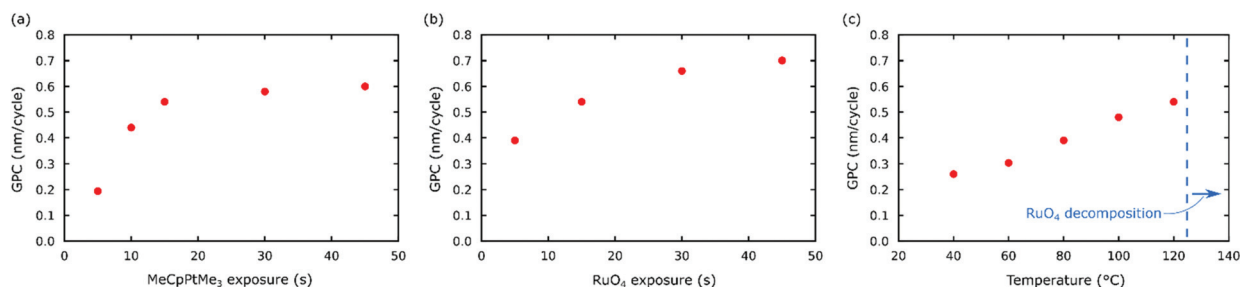


Fig. 5 (a) Growth per cycle determined by *ex situ* XRR as a function of the MeCpPtMe₃ precursor exposure time for a fixed RuO_4 pulse time of 15 s, at 120 °C sample temperature. As the MeCpPtMe₃ precursor pulse is of a static nature, 30 s and 45 s exposures correspond to two and three successive static MeCpPtMe₃ pulses of 15 s, respectively, while the 5 s and 15 s exposures correspond to single static pulses. (b) Growth per cycle determined by *ex situ* XRR as a function of the RuO_4 co-reactant exposure time for a fixed MeCpPtMe₃ precursor pulsing time of 15 s, at 120 °C sample temperature. As the RuO_4 pulse is of a static nature, 30 s and 45 s exposures correspond to two and three successive static RuO_4 pulses, respectively, while the 5 s, 10 s and 15 s exposures correspond to single static pulses. (c) Growth per cycle as a function of the substrate temperature for the MeCpPtMe₃/RuO₄ process with 15 s exposure for both MeCpPtMe₃ and RuO_4 .

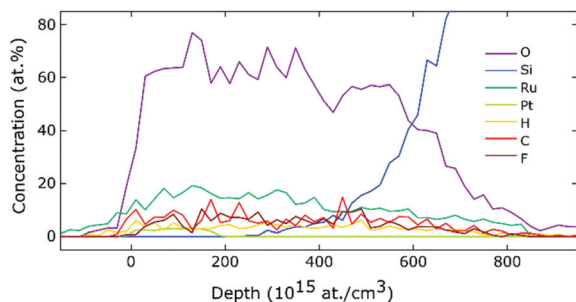


Fig. 6 Elemental distribution obtained from analysis of ERD spectra acquired for a ca. 60 nm film grown with the MeCpPtMe₃/RuO₄ ALD process.

The elements detected in XPS were platinum, ruthenium and oxygen (Fig. S2, ESI†), while the carbon content in the films could not be determined due to the overlap of the C 1s and Ru 3d^{3/2} peaks. Evaluating the peak intensities of the Pt 4f and Ru 3p signals corroborates a Ru-rich composition (1 : 5.3 ratio of Pt to Ru in the bulk of the film). The depth profile extracted from ERD analysis of a ca. 60 nm film is depicted in Fig. 6. Pt, Ru and O concentrations of respectively 2.51 ± 0.06 at%, 16.2 ± 0.5 at% and 64 ± 2 at% were extracted, while ERD also provides the concentrations of the light element impurities. The carbon and hydrogen content was slightly lower than for the TMA/RuO₄ deposited thin film, *i.e.* 7.5 ± 0.9 at% and 3.5 ± 0.4 at%, respectively. Unexpectedly, an opposite trend is observed for the fluorine contamination. While for the TMA/RuO₄ process trace amounts below 1 at% were found in the film, about 6.0 ± 0.7 at% of fluorine was incorporated in the film grown with the MeCpPtMe₃/RuO₄ process.

Overall, the results obtained for the MeCpPtMe₃/RuO₄ process show promise that the RuO₄-based ALD-route presented here can be applicable to a wide variety of metal precursors, yielding MRuO_x or M-doped RuO_x thin films. The Ru-rich nature of the deposited thin films may be related to the specific ligands of the Pt precursor molecule. If several RuO₄ molecules react with the remaining ligands of a single adsorbed Pt precursor molecule at the surface, this can explain the high proportion of Ru relative to Pt in the resulting films as well as the high growth rate observed for the process, *i.e.* 0.52 nm per cycle at 120 °C. More insights in the fundamentals of the surface chemistry could provide hints on how one can tune the M to Ru ratio by precursor selection.

Conclusions

In summary, we have reported a low temperature (<125 °C) ALD route to amorphous metal ruthenates based on the use of RuO₄ as the co-reactant, here exemplified for aluminium ruthenate and platinum ruthenate. Both the TMA/RuO₄ and MeCpPtMe₃/RuO₄ process exhibit self-limiting surface reactions and linear growth as a function of the number of ALD cycles. The observed saturated GPCs (0.86 nm per cycle and

0.52 nm per cycle for the aluminium and platinum ruthenate process, respectively) are relatively high compared to common values for binary ALD processes. We propose that RuO₄ combusts the organic ligands of the adsorbed precursor molecules, while at the same time binding a substantial amount of RuO₂ to the surface. The presence of hydrogen and carbon impurities (both *ca.* 9% in the aluminium ruthenate films) nevertheless indicates that some ligand fragments can be incorporated in the film. More in-depth studies on the contamination levels and reaction mechanism of TMA and MeCpPtMe₃ with a RuO₂ surface are planned to understand the origin of the particular GPC values and metal (Al, Pt) to Ru ratios obtained in these processes.

Author contributions

Matthias M. Minjauw: Conceptualization, investigation, formal analysis, writing: original draft. Ji-Yu Feng: Investigation, formal analysis, writing: original draft. Timo Sajavaara: Investigation, formal analysis. Christophe Detavernier: Conceptualization, resources, supervision, writing: review and editing. Jolien Dendooven: Conceptualization, resources, supervision, writing: review and editing.

Conflicts of interest

There are no conflicts to declare.

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